A Spectrum of Models for Low-Level Feedback Control Design of Small Underactuated Uncrewed Surface Vehicles

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***Abstract*—Effective low-level feedback control of small underactuated uncrewed surface vehicles (USVs), specifically vessels under seven meters with a single thruster and rudder, requires dynamic models that provide actionable design insight, not predictive accuracy alone. Tow-tank testing is impractical at this scale; parameters must be identified from limited field trials. Existing system identification literature largely pursues model fidelity as an end in itself, producing representations whose complexity can obscure which aspects of the dynamics are structurally significant for compensator design.**

We propose a methodology built around a deliberate spectrum of three model levels, each identified from the same experimental data: (1) first-order, linear, single-input single-output (SISO) transfer functions, Nomoto-style models for surge and yaw that are immediately interpretable and support controller architecture intuition, but risk oversimplification when considered in isolation; (2) a physics-based nonlinear state-space formulation incorporating quadratic hydrodynamic drag; and (3) a Nonlinear AutoRegressive with eXogenous inputs (NARX) multilayer perceptron. Rather than seeking a single best model, the spectrum is used collectively: simple models anchor human intuition; structured nonlinear models expose which terms are dynamically salient; and the black-box model bounds the benefit of added complexity.

Preliminary results from field step-response experiments demonstrate the approach. First-order SISO models achieve R2=0.93 (surge, tau=1.95 s) and R2=0.83 (yaw rate, tau=0.47 s), providing direct controller tuning parameters. Cross-model diagnostics identify quadratic drag as the dominant surge nonlinearity and reveal that speed-dependent rudder effectiveness, a key coupling for underactuated control design, is not resolvable from a single short trial, motivating targeted data collection. The NARX model matches one-step prediction accuracy (R2=0.93) but diverges in closed-loop simulation, demonstrating that physical structure provides essential regularization under limited data. A comprehensive dataset spanning multiple operating regimes is being collected to evaluate the full methodology.

***Index Terms*—uncrewed surface vehicle, system identification, model hierarchy, feedback control design, nonlinear dynamics, data-driven modeling, underactuated vessels**

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References

[1] G. E. P. Box and N. R. Draper, *Empirical Model Building and Response Surfaces*. Wiley, 1987, source of the aphorism "all models are wrong, but some are useful".
[2] T. I. Fossen, *Handbook of Marine Craft Hydrodynamics and Motion Control*. Wiley, 2011.
[3] K. Nomoto, T. Taguchi, K. Honda, and S. Hirano, "On the steering qualities of ships," *International Shipbuilding Progress*, vol. 4, no. 35, pp. 354–370, 1957.
[4] C. R. Sonnenburg and C. A. Woolsey, "Modeling, identification, and control of an unmanned surface vehicle," *Journal of Field Robotics*, vol. 30, no. 3, pp. 371–398, 2013, compares model hierarchy from 3-DOF nonlinear to Nomoto first-order; finds first-order steering model best compromise at higher speeds.
[5] ——, "Control-oriented planar motion modeling of unmanned surface vehicles," in *OCEANS 2010 IEEE – Sydney*, 2010.
[6] N. Wang *et al.*, "Modeling and identification of an unmanned surface vehicle based on sea trials data," in *2020 Chinese Automation Congress (CAC)*, 2020, 3-DOF nonlinear model identified from sea trials for control.
[7] W. Caharija *et al.*, "An approach for system identification in unmanned surface vehicles," in *OCEANS 2017 – Aberdeen*, 2017.
[8] J. Woo, J. Park, C. Yu, and N. Kim, "Dynamic model identification of unmanned surface vehicles using deep learning network," *Applied Ocean Research*, vol. 78, pp. 123–133, 2018, lSTM-RNN reduces surge error 76.9% and yaw-rate error 60.7% over linear models; no closed-loop simulation evaluation.

[9] P. Xu, C. Han, H. Cheng, C. Cheng, and T. Ge, “A physics-informed neural network for the prediction of unmanned surface vehicle dynamics,” *Journal of Marine Science and Engineering*, vol. 10, no. 2, p. 148, 2022, pINN combining physical 3-DOF structure with data-driven learning; better generalization than pure NN with small training sets.

[10] A. Tanveer and S. M. Ahmad, “Unmanned surface vehicle: Yaw modeling and identification,” arXiv:2303.13064, 2023, linear second-order yaw model identified from pool trials. [Online]. Available: https://arxiv.org/abs/2303.13064

[11] E. I. Sarda, H. Qu, I. R. Bertaska, and K. D. von Ellenrieder, “Station-keeping control of an unmanned surface vehicle exposed to current and wind disturbances,” in *OCEANS 2016 MTS/IEEE Monterey*, 2016.

[12] Z. Liu, Y. Zhang, X. Yu, and C. Yuan, “Unmanned surface vehicles: An overview of developments and challenges,” *Annual Reviews in Control*, vol. 41, pp. 71–93, 2016.